

Ibn Tofail University

Rigid Body Mechanics

Problem Set 1

Course Question*

Show that any antisymmetric vector field is equiprojective and conversely.

Correction

Exercise 1: Vector Division

Solve the vector equation $\vec{U} \times \vec{X} = \vec{V}$ where $\vec{U}$ and $\vec{V}$ are two non-zero vectors.

Correction

Exercise 2:

Let $[T]$ be a torsor defined by its reduction elements at point P: $\vec{R}$ and $\vec{m}(P)$. We consider the points O(0, 0, 0), A(1, 0, 0) and B(1, 1, 0) in a Cartesian reference frame $R(O, \vec{i}, \vec{j}, \vec{k})$. Knowing that we have certain components of the moments at O, A and B namely:

$$\vec{m}(O) = \begin{pmatrix} \alpha \\ \alpha \\ 1 \end{pmatrix}, \quad \vec{m}(A) = \begin{pmatrix} x \\ 1 - \alpha \\ 2 \end{pmatrix} \quad \text{and} \quad \vec{m}(B) = \begin{pmatrix} y \\ z \\ -\alpha \end{pmatrix}$$

1) Completely determine $\vec{m}(A)$ and $\vec{m}(B)$ by finding the values of x, y, and z as functions of α .

2) Calculate the resultant of the torsor $[T]$ associated with this moment field.

3) Find the central axis of the torsor $[T]$.

4) Can the vector system that generates this field be equivalent to a couple torsor?

Correction

Exercise 3:

Prove that two torsors $[T_1]$ and $[T_2]$ are equal if, and only if, there exist three non-aligned points P_1 , P_2 and P_3 at which their moments are equal.

Correction**Exercise 4:**

Let [T] be the torsor associated with the field of sliders defined on the domain (D) consisting of four points M_1 , M_2 , M_3 and M_4 such that the resultants associated with each point are:

$$\begin{aligned} M_1(2, -2, 3) &\rightarrow \vec{R}_1(5, 0, 0) \\ M_2(-4, 2, -1) &\rightarrow \vec{R}_2(0, -2, 0) \\ M_3(5, -2, 3) &\rightarrow \vec{R}_3(0, 0, 3) \\ M_4(0, 2, 0) &\rightarrow \vec{R}_4(3, 4, 1) \end{aligned}$$

- 1) Determine the resultant of the torsor and its moment at point O.
- 2) Calculate the scalar invariant of the torsor and conclude.
- 3) Deduce the moment of the torsor at any point P.
- 4) Find the equations of its axis.

Correction**Exercise 5:**

Consider, in a Cartesian reference frame $R(O, X, Y, Z)$ with basis $(\vec{i}, \vec{j}, \vec{k})$, the system of three bound vectors defined by:

$$\begin{aligned} \vec{V}_1 &= a\vec{i} + b\vec{j} + \vec{k} \quad \text{with origin } A_1(1, 0, 0) \\ \vec{V}_2 &= 2b\vec{i} - 2a\vec{j} - 3\vec{k} \quad \text{with origin } A_2(0, 1, 0) \\ \vec{V}_3 &= -8\vec{i} + \vec{j} + c\vec{k} \quad \text{with origin } A_3(0, 0, 1) \end{aligned}$$

- 1) For which values of a, b and c is the system of these three bound vectors equivalent to a couple torsor?
- 2) Then determine its moment at point O.

Correction